INTERNSHIP MONTHLY REPORT

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Details of the internship schedule:

Key date	Key learning activities
Week 9 (3/9 - 5/9)	Get familiar with the problem of localization using Apriltag
Week 10 (8/9 - 12/9)	Compute the position in space of an object based on Apriltag visual input
Week 11 (15/9-19/9)	Simulation of precision landing using Apriltag
Week 12 (19/9 - 26/9)	Precision landing and navigation in an indoor non-GPS environment using computer vision

1. **What skills have you learned in the past 4 weeks? Please describe your progress using the suggestions below:**

Over the past 4 weeks, I have learned several valuable skills related to computer vision and navigation systems. In Week 9, I became familiar with the problem of localization using AprilTag, a visual marker system that enables accurate tracking and positioning. By Week 10, I was able to compute the position of an object in space based on AprilTag visual input, enhancing my understanding of how to extract real-time spatial information from visual data. In Week 11, I worked on simulating precision landing using AprilTag, which involved improving the accuracy of landing mechanisms based on visual feedback. Finally, in Week 12, I focused on precision landing and navigation in an indoor, non-GPS environment using computer vision techniques, which provided me with a deeper understanding of how to navigate in GPS-denied spaces. These experiences have strengthened my skills in localization, computer vision, and autonomous navigation, which are critical in robotics and drone applications.

One of the biggest challenges I faced was working with matrix and frame transformations in 3D space, especially when merging data from different coordinate frames. In robotics and computer vision, it's essential to align information from various sensors or reference points into a common frame of reference, which requires transforming between coordinate systems. I had to deal with the complexity of applying rotation matrices, translation vectors, and homogeneous transformations to accurately translate and rotate points between different frames. This process was particularly challenging when working with AprilTag localization, as it involved understanding how the visual input from the tags needed to be transformed to match the robot's or drone's coordinate system.

2. **Are there any problems or delays with the internship where you need support?**

Everything ended well and I feel that myself has gain alot of new knowledge and experience.

With respect to the final report, with questions on professional responsibilities and on consideration of standards and laws:

When designing and deploying systems that rely on visual markers like AprilTags, it is important to consider privacy laws (e.g., GDPR) to ensure that no personal data is inadvertently collected or misused. Moreover, professionals should remain informed about any legal restrictions or emerging regulations related to autonomous systems, particularly in scenarios where technology interacts with public spaces or people, such as in drones or robotic prosthetics. By recognizing these legal and ethical considerations, engineers can ensure that their work contributes positively to society, promotes safety, and avoids potential legal complications.